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Physiological Data Collected from Wearable Devices Identify and Predict Inflammatory Bowel Disease Flares

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Abstract

Background and Aims: Wearable devices capture physiological signals non-invasively and passively. Many of these parameters have been linked to inflammatory bowel disease (IBD) activity. We evaluated the associative ability of several physiological metrics with IBD flares and how they change before the development of flare.

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Conflicts of Interest:

All other authors declare no relevant conflicts of interest.

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Methods: Participants throughout the United States answered daily disease activity surveys and wore an Apple Watch, Fitbit or Oura Ring. These devices collected longitudinal heart rate (HR), resting heart rate (RHR), heart rate variability (HRV), steps, and oxygenation (SpO₂). C-reactive protein, erythrocyte sedimentation rate and fecal calprotectin were collected as standard of care. Linear mixed-effect models were implemented to analyze HR, RHR, steps, and SpO₂, while cosinor mixed-effect models were applied to HRV circadian features. Mixed effect logistic regression was used to determine the predictive ability of physiological metrics.

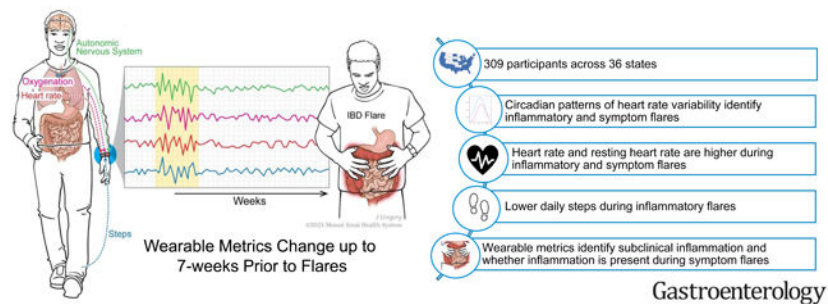
Results: 309 participants were enrolled across 36 states. Circadian patterns of HRV significantly differed between periods of inflammatory flare and remission, and symptomatic flare and remission. Marginal means for HR and RHR were higher during periods of inflammatory flare and symptomatic flare. There was lower daily steps during inflammatory flares. HRV, HR, and RHR differentiated whether participants with symptoms had inflammation. HRV, HR, RHR, steps, and SpO₂ were significantly altered up to 7 weeks prior to inflammatory and symptomatic flares.

Conclusions: Longitudinally collected physiological metrics from wearable devices can identify and change prior to IBD flares, suggesting their feasibility to monitor and predict IBD activity.

Lay Summary:

Commonly used commercially available wearable devices can identify and predict inflammatory bowel disease exacerbations opening the potential for their use to monitor the disease.

Graphical Abstract



Keywords

Wearable Devices; inflammatory bowel disease; Prediction; Crohn's disease; ulcerative colitis

INTRODUCTION

The inflammatory bowel diseases (IBD), comprised of ulcerative colitis (UC) and Crohn's disease (CD), are chronic relapsing and remitting inflammatory diseases of the gastrointestinal tract.^{1,2} Its management is challenging due to unpredictable and frequent disease flares, and frequent discordance of inflammation and symptoms.³ Current modalities of disease monitoring have significant limitations that rely on patient reported symptoms and the assessment of inflammation using blood or stool testing, imaging or ileocolonoscopy. However, evaluations are limited to a single time point of assessment, are inconvenient, and

can be invasive. New monitoring modalities are needed which can assess disease activity longitudinally, passively and in real-time.

Wearable devices are used by approximately 20% of individuals in the United States.⁴ They can measure in a near continuous or continuous manner physiological metrics including heart rate (HR), activity, resting heart rate (RHR), and heart rate variability (HRV). Wearable devices are an increasingly accepted tool for monitoring health and disease. They are frequently employed in non-inflammatory based diseases for remote patient monitoring, allowing individuals to be monitored outside of the clinical setting, which have resulted in improved outcomes in multiple disease states.^{5,6} There is increasing interest in the monitoring of chronic inflammatory diseases and the prediction of flares and medication response. There are limited publications in this space due to the difficulty in monitoring inflammatory based conditions remotely via wearable technology. Individuals with IBD are interested in using wearable technology to manage their disease.⁷ Yet, there are limited studies published that evaluate wearable technology in relation to IBD activity. One small study followed 37 patients with IBD after bowel surgery, finding that activity and sleep data from wearable biosensors predicted the postoperative length of hospital stay.⁸ Two studies evaluating a sweat-sensing wearable device further demonstrated that sweat-based assessments of immune and inflammatory markers correlate with serum measurements. These studies, while small, demonstrate the feasibility of passively monitoring markers of disease activity in IBD.^{9,10}

Commonly used wearable devices can assess physiological signals which are altered in chronic inflammatory diseases, thereby creating an opportunity for disease monitoring. Chronic inflammatory diseases, such as IBD, are associated with impaired autonomic nervous system (ANS) function. Both UC and CD have been found to have an impaired cardiovagal tone, and a relative parasympathetic suppression and sympathetic predominance.^{11–13} ANS alterations are associated with changes in several physiological metrics which can be measured with wearable technology, providing an opportunity to monitor inflammatory diseases using these tools. An important parameter that can be measured by wearable technology is HRV. HRV is a measure of the small-time differences between each heartbeat. Changes in the beat-to-beat heartrate are controlled by both sympathetic (SNS) and parasympathetic nervous system (PNS) activity. HRV is an indirect measure of the ANS.¹⁴ Higher HRV reflects a higher PNS tone, while lower HRV reflects higher SNS tone. Cross-sectional studies have demonstrated a decrease in HRV in IBD. Additionally they have shown that parasympathetic function displays a stronger inverse association with disease activity compared to sympathetic function supporting overall ANS dysfunction in IBD.¹⁵ Similarly, HR is affected by changes in ANS function with increased sympathetic tone raising HR.¹⁶

To demonstrate the feasibility of wearable devices to monitor physiological metrics and their relationship to IBD disease activity, we performed a pilot study where 15 participants with UC wore a wearable patch to measure HRV over 9 months. We found that longitudinally collected increased SNS activity was associated with UC symptoms and elevated inflammatory markers.¹⁷ Our current study builds upon this initial observation. To

evaluate the ability of multiple physiological signals collected from wearable devices to identify IBD flares we launched the IBD Forecast Study.

MATERIALS AND METHODS

Study Population

The IBD Forecast Study is a prospective exploratory observational cohort study enrolling individuals throughout the United States. Eligible participants (1) have a diagnosis of CD or UC, (2) are ≥ 18 years of age, (3) are on a medication to treat IBD, and (4) are willing to use a wearable device. Participants were excluded if they (1) are pregnant, (2) have significant medical conditions such as heart failure or another chronic disease, (3) have a pacemaker or defibrillator, or (4) use medications that impact HR or HRV such as beta blockers, calcium channel blockers, or benzodiazepines.

Participants were nationally recruited via social media advertising, study flyers, a study website, and an email about the study to individuals with IBD who visited the Inflammatory Bowel Disease Center at Mount Sinai Hospital, USA. A participant could use their own Apple Watch, Fitbit, or Oura Ring, or be provided with an Apple Watch or Fitbit for the duration of the study. No compensation was provided. Participants were able to contribute for as long as they would like. The study opened in December 2021 and closed in June 2023.

Study Procedures

Participants downloaded ehive, our custom digital research platform, to their smartphones, self-verified inclusion and exclusion criteria and signed the electronic study consent.¹⁸ Participants linked their wearable device or requested a wearable device from the study team, provided baseline demographic information, medical and surgical history, and IBD history through the app. Participants were recommended to wear their wearable device for a minimum of 8 hours per day. Participants who used their wearable device or answered daily surveys less than 4 days per week were sent a smart phone push notification reminding them to participate.

Wearable Devices and Physiological Metrics

This study utilized four physiological metrics from three device types: Apple Watch, Fitbit, and Oura Ring. The metrics that were collected by all the devices included HRV, HR, RHR, and daily steps. These wearable devices are commercially available and contain photoplethysmography optical sensors. This enables the assessment of capillary volume changes, the determination of the number of heart-beats and calculation of the time between heart beats for HRV measurements.^{19–21} HR is calculated as beats per minute (BPM) throughout the day and night and RHR is defined as an estimation of a user's lowest HR's during periods of rest. The Apple Watch calculates the standard deviation of the IBI of normal sinus beats (SDNN) which is a time domain HRV metric that reflects both sympathetic and parasympathetic nervous system activity. Lower values reflect an increased sympathetic tone. The Fitbit and Oura Ring calculate HRV in the time domain metric root mean square of successive differences between normal heartbeats (RMSSD). The RMSSD is more influenced by the PNS compared to the SNS, with lower values reflecting increased

sympathetic tone.²² The Apple Watch Series 6 or later can measure the oxygen saturation of arterial hemoglobin (SpO2) using conventional pulse oximetry methods. This feature has been validated in several studies.²³ See Supplemental Methods for further information.

Longitudinal Clinical and Laboratory Assessment

Clinical disease activity was assessed using baseline and daily UC or CD specific Patient Reported Outcome (PRO-2) scores. The CD specific PRO-2 is derived from the CD Activity Index and asks participants to rate their abdominal pain and stool frequency with the components weighted to create a cumulative score. The UC specific PRO-2 is derived from the rectal bleeding and stool frequency sub-scores of the Mayo Clinic Score for UC. It consists of two questions that ask participants to rate their degree of rectal bleeding and stool frequency. C-reactive protein (CRP), erythrocyte sedimentation rate (ESR) and fecal calprotectin (FC) results were collected as part of each participant's standard of care management. Results were reported by participants in ehive. Laboratory values that were not reported were captured through the participants' electronic health record, if he or she was followed at Mount Sinai Hospital.

Symptomatic and Inflammatory Flare Criteria

IBD flares were characterized as being consistent with inflammatory flares, denoted by elevated inflammatory biomarkers, or symptomatic flares, according to the presence of symptoms consistent with an IBD flare. In each timeframe, patients could meet the study definition of (1) inflammatory flare either with or without symptomatic flare, (2) symptomatic flare without inflammation or (3) remission. Inflammatory flare was defined by elevation of CRP, ESR or FC values. An FC >150 µcg/g was consistent with active inflammation.^{24,25} A CRP and ESR value was considered elevated if they were >5 mg/dL and >30 mm/hour, respectively.²⁶ We assumed that the underlying inflammatory state extended for a period before and after the laboratory test is obtained. To account for this, and to avoid a misclassification of inflammatory status around each laboratory test, CRP, ESR and FC values were inputted in a window of +/-7 days. For the CD PRO-2, remission was defined as a PRO-2 score <8.^{27,28} In UC symptomatic remission was defined using a PRO-2 score ≤ 1 with a rectal bleeding score = 0 and stool frequency score ≤ 1.²⁹

There is no consensus for the minimum number of daily symptom-based surveys denoting symptomatic activity required to classify a symptomatic flare. To evaluate potential criteria, we performed a sensitivity analysis to assess several definitions of symptomatic disease flare described in detail in Supplemental Figure 1. Based on this analysis, we limited our definition of symptomatic flare to having at least 4 surveys answered in each 7-day period, and with at least 2 of these surveys meeting the criteria for symptomatic flare. This was consistent with our a priori definition of study compliance, which asked participants to answer a minimum of 4 surveys per 7-day period.

Statistical Analysis

Sex, age, device type and Body Mass Index (BMI) data were utilized as covariates in all statistical analyses. No association was found between patients with reported or missing demographic information, supporting no departure from missing at random mechanism. As

such, missing values were imputed based on the multivariate distribution of the observed values using the function `gImpute` from `Hmisc` R package.³⁰ Longitudinal wearable measures were not imputed for this analysis. We use a mixed-model approach that produces robust estimates when missing values in the wearable data collection are missing at random. Factors leading to missing wearable data are dictated by technological and behavioral factors, not easily determined. However, considering the large data set, comparisons made within individuals, and models using wearable data collected when clinical information is recorded, the missing at random (MAR) assumption is reasonably met.

HRV Analysis

Longitudinal changes in HRV have been shown to have daily circadian patterns over a 24-hour period and can be modeled using a non-linear Cosinor model approach.³¹ Since our analysis evaluates the changes in the daily pattern as a function of flares in a longitudinal study, those changes were evaluated using Cosinor mixed effects models. The analyses were carried out using the `CosinorRmixedeffects` R package,³² which is publicly available and was developed by our team. 500 bootstraps were applied. This non-linear model, is fully described by Hou et al.³² This approach models the HRV circadian rhythm over 24 hours. This pattern can be described using the circadian parameters: (1) MESOR: a rhythm-adjusted mean; (2) Amplitude: a measure of half the extent of variation within a day; and (3) Acrophase: a measure of the time that overall high values occur each day.³³ The model included inflammatory/symptomatic flare status as well as age, sex, and BMI as covariates. Device type was included as a fixed effect. Further details are available in the Supplemental Methods.

HR, RHR, SpO2 and Steps Analysis

Linear mixed-effect models were fitted to analyze the physiological parameters HR, RHR, and steps for all devices, and SpO2 solely for the Apple Watch. Metrics were analyzed on an hourly basis (HRV, SpO2, HR) and daily basis (RHR, steps). Models include fixed effects for covariates age, sex, and BMI, along with inflammatory and symptomatic flare predictors (and its interaction for Figure 2), as well as a random intercept for each participant for per-day models, and a random intercept nested by day for all per-hour models. These models were fitted using the `lme` function from the `nlme` R package.³⁴ Furthermore, heterogeneous variances within flare/remission stage were incorporated into our modelling with the final model defined in terms of LRT/AIC, as described above. Marginal means with 95% confidence intervals were estimated using the `emmeans` package,³⁵ as well as testing the hypothesis of interests, namely differences between flare stages, through contrasts. To evaluate the effect of the presence or absence of nocturnal or daytime wearable data on outcomes we performed a sensitivity analysis. RHR collected at night versus RHR collected over 24 hours did not result in significant differences (results not presented) in outcomes, demonstrating the negligible effect of the presence or absence of nocturnal data.

Physiological Parameters as an Early Signal of Flares

Mixed-effect logistic regression was utilized to evaluate the temporal association of various physiological parameters, or their combination, in determining the probability of experiencing flare-ups on the same day or before the flare-up periods occurs. Specifically,

for each flare definition, we evaluated the changes in the physiological parameter taken at 7, 14, 21, 28, 35, 42, and 49 days before inflammatory or symptomatic flares. To evaluate models' performance, specificity, sensitivity, accuracy, precision, recall, area under precision-recall curve (AUPRC), AUC and the F1-score were estimated. Further details are available in the Supplemental Methods.

RESULTS

Cohort Characteristics

A total of 309 participants consented and contributed both wearable device and survey data across 36 states within the United States. Participants mean age was 39.9 (SD 14.4) years old, with 208 (67.3%) being female. 196 participants had CD and 113 subjects had UC. The mean duration of follow up was 213 (SD 156) days. 255 participants wore an Apple Watch, 53 participants wore a Fitbit while 16 participants wore an Oura Ring, as some subjects used 2 types of devices over the study period (Table 1). In total, 120 devices were loaned to participants (107 Apple Watches, 13 Fitbits). There were 152 days in inflammatory flare and 999.7 7-day periods defined as symptomatic flares. Symptomatic and inflammatory trajectories of each participant can be visualized in Supplemental Figure 2. Mean wearable device usage statistics are presented in Supplemental Table 1.

Identification of Inflammatory Flares

HRV differentiated inflammatory flares compared to inflammatory remission, with significant differences in the circadian pattern of HRV measurements observed across device types (Figure 1a). The MESOR of the circadian pattern of SDNN was lower (39.26 (36.69–42.10) ms) during an inflammatory flare compared to periods of inflammatory remission (41.06 (38.65–43.80) ms; $p < 0.0001$). There were no differences in the Acrophase ($p = 0.40$), Amplitude ($p = 0.87$) or time to peak ($p = 0.40$) of the circadian measurements. There was a significant difference in the MESOR (38.31 (32.63–44.58) ms; $p = 0.03$) of the circadian pattern of RMSSD in participants in an inflammatory flare compared to those without inflammation (MESOR 35.87 (29.88–42.41) ms). There was no difference in the Acrophase ($p = 0.07$), Amplitude ($p = 0.15$) or Peak Time ($p = 0.07$).

Physiological data differentiated inflammatory periods from non-inflammatory periods in participants with IBD (Figure 1b). The mean HR and RHR were higher during inflammatory flares (HR 79.31 ± 0.92 (mean $\pm$ standard error of the mean, SEM); RHR 65.30 ± 0.80) and daily steps lower (steps $5,082 \pm 0.02$) compared to periods of inflammatory remission (HR 78.92 ± 0.92 , $p = 0.03$; RHR 64.64 ± 0.78 , $p = 0.02$, steps $5,507 \pm 0.02$, $p = 0.01$). The mean SpO₂ was not different during inflammatory flares ($96.52\% \pm 0.14$) compared to periods of inflammatory remission ($96.49\% \pm 0.13$; $p = 0.61$).

Identification of Clinical Flares

Inflammatory Bowel Disease: Physiological metrics differentiate symptomatic flares from symptomatic remission in those with IBD. In participants with IBD, including both CD and UC, there were significant differences in the circadian pattern of HRV measurements across devices and measurement types. The Amplitude of the circadian pattern of SDNN

was higher (4.68 ms, 95%CI 3.87–5.60) during symptomatic flares compared to periods without symptoms (3.69 ms, 95%CI 2.96–4.51, $p=0.006$) (Figure 1c). There was no difference in the Acrophase ($p=0.66$), MESOR ($p=0.26$) or Peak Time ($p=0.66$) of the circadian pattern of HRV. There were no differences observed in the circadian pattern of RMSSD during symptomatic flares compared to periods without symptoms (MESOR, $p=0.81$; Amplitude, $p=0.96$; Acrophase, $p=0.96$; Peak Time, $p=0.96$).

The mean HR and RHR were higher during symptomatic flares (HR 78.32 ± 0.68 ; RHR 64.60 ± 0.62) compared to periods of symptomatic remission (HR 77.89 ± 0.68 , $p<0.0001$; RHR 64.11 ± 0.61 , $p=0.0009$) (Figure 1b). Daily steps and SpO2 were not different during symptomatic flares (steps $5,564\pm0.02$, SpO2 $96.50\%\pm0.11$) compared to periods in symptomatic remission (steps $5,649\pm0.02$, $p=0.32$, SpO2 $96.55\%\pm0.10$; $p=0.10$).

Crohn's Disease

In participants with CD there were no differences in the circadian pattern of SDNN (MESOR $p=0.83$, Amplitude $p=0.44$, Acrophase $p=0.40$, Peak Time $p=0.40$) or RMSSD (MESOR, $p=0.06$; Amplitude, $p=0.38$; Acrophase, $p=0.65$; Peak Time, $p=0.65$) during a symptomatic flares compared to symptomatic remission. Daily steps were not different between symptomatic flares of CD compared to periods of symptomatic remission (steps $5,193\pm0.02$ vs steps $5,385\pm0.02$, respectively; $p=0.11$). There was no significant difference in HR (78.10 ± 0.92 vs 78.08 ± 0.91 , respectively; $p=0.88$) and RHR (64.69 ± 0.85 vs 64.55 ± 0.84 ; $p=0.58$) between the two states. However, SpO2 was significantly lower during symptomatic flares compared to periods of symptomatic remission ($96.44\%\pm0.14$ vs $96.63\%\pm0.14$, respectively; $p=0.0002$) between the two states (Figure 1b).

Ulcerative Colitis

In participants with UC, the circadian pattern of SDNN significantly differed between periods of symptomatic flare compared to periods of remission. There was a significant difference in the Amplitude ($p=0.002$) of the circadian pattern of SDNN between periods of symptomatic flares (5.30 (4.07–6.67) ms) and symptomatic remission (3.76 (2.81–4.97) ms), while there were no differences seen in the other circadian features of HRV (MESOR $p=0.24$, Acrophase $p=0.89$, Peak Time $p=0.89$). Differences in the circadian patterns of RMSSD were able to differentiate periods of symptomatic flare (MESOR, 37.34 (28.71–46.06) ms) compared to symptomatic remission (34.52 (25.90–43.16) ms; $p<0.0001$). Daily steps did not differ during symptomatic flares of UC compared to periods of symptomatic remission (steps $6,103 \pm 0.02$ vs steps $6,079 \pm 0.02$ respectively; $p=0.85$). There was a significant increase in HR (78.89 ± 1.04 vs 78.20 ± 1.04 , respectively; $p<0.0001$) and RHR (64.53 ± 0.92 vs 63.81 ± 0.91 ; $p=0.0001$) during symptomatic flares compared to periods of symptomatic remission. There was no difference in SpO2 ($96.56\% \pm 0.16$ vs $96.53\% \pm 0.16$, respectively; $p=0.40$) between periods of symptomatic flare and remission (Figure 1b).

Interactions Between Symptoms and Inflammation

We sought to determine whether physiological metrics collected from wearable devices can differentiate the presence and absence of symptoms and underlying inflammation. The MESOR ($p<0.0001$), Acrophase ($p<0.0001$) and Peak Time ($p<0.0001$) of SDNN

significantly differed between periods of symptomatic and inflammatory flare compared to periods with symptomatic flare but no inflammation (Supplemental Table 2), thus identifying whether inflammation is present or absent during symptomatic periods. Additionally, there was a significant difference in the MESOR ($p < 0.0001$), Acrophase ($p = 0.03$) and Peak Time ($p = 0.03$) of the circadian pattern of SDNN during periods with and without symptomatic flares in the presence of underlying inflammation. There were no significant differences seen in the circadian pattern of SDNN during periods in symptomatic remission and underlying inflammation compared to periods in both symptomatic and inflammatory remission. There were no differences in the MESOR, Amplitude, Acrophase or Peak Time of the circadian pattern of SDNN in participants with a symptomatic flare compared to those in symptomatic remission, if there was no underlying inflammatory flare ($p > 0.05$) (Figure 2a).

Similar findings were observed when evaluating other physiological metrics collected from the wearable devices. Periods with symptomatic flares and inflammatory flares compared to periods with symptomatic flares but in inflammatory remission were differentiated by RHR (67.20 ± 1.09 vs 65.28 ± 0.96 respectively; $p = 0.006$) and HR (81.08 ± 1.02 vs 78.43 ± 1.00 respectively; $p < 0.001$), but not SpO2 ($96.35\% \pm 0.20$ vs $96.13\% \pm 0.17$, respectively; $p = 0.07$). Comparing periods of symptomatic remission and inflammatory flare to periods with no symptoms and no inflammation, there was no difference in RHR and HR (64.60 ± 0.96 vs 64.40 ± 0.92 , respectively, $p = 0.62$; 77.57 ± 1.00 vs 77.64 ± 0.99 , respectively; $p = 0.82$). However, SpO2 was able to differentiate these two states ($96.26\% \pm 0.19$ vs $96.52\% \pm 0.17$, respectively; $p = 0.04$). RHR, HR, and SpO2 were able to differentiate periods of symptomatic flare compared to symptomatic remission if there was no inflammation present (65.28 ± 0.96 vs 64.40 ± 0.92 , $p = 0.03$; 78.43 ± 1.00 vs 77.64 ± 0.99 , $p = 0.003$; $96.13\% \pm 0.17$ vs $96.52\% \pm 0.17$; $p < 0.0001$, respectively) (Figure 2b).

Change Preceding Inflammatory and Symptom Flares

Physiological metrics were evaluated for their changes 7, 14, 21, 28, 35, 42, and 49 days prior to flares of IBD (Figure 3a). Analysis was limited to Apple Watch-derived metrics to enable direct comparison of the changes in measured metrics before flares. HRV (SDNN), HR, RHR, Steps and SpO2, as well as models combining these metrics, maintained high F1 scores throughout the 49 day period prior to inflammatory and symptomatic flares (Figure 3b). F1 scores balance precision and recall and are optimal for evaluating imbalanced datasets. When evaluating the physiological parameters that were analyzed on a per day basis (HR, Steps, RHR), a model including all metrics maintains an F1 score of 0.88 at 49-days preceding identification of inflammatory flares (49 days before flare; AUC 0.98 [0.97–0.99]; F1 0.88; AUPRC 0.55; Sensitivity 0.92; Specificity 0.94) (Supplemental Figure 3a and 3b, Supplemental Table 3). Similar results were found when evaluating physiological parameters analyzed on a per hour basis (HRV, HR, SpO2). A model including all these metrics has an F1 score of 0.90 at 49-days preceding identification of inflammatory flares (49 days before flare; AUC 0.99 [0.99–0.99]; F1 0.90; AUPRC 0.60; Sensitivity 0.91; Specificity 0.97) (Supplemental Table 4). The F1 scores for symptomatic flares were similarly high and largely stable over the 49 days prior to flare (Supplemental Figure 3a and 3b). A model including the parameters measured on a per day basis (HR, Steps, RHR), had

an F1 score of 0.83 at 49-days preceding the flare event (49 days before flare; AUC 0.96 [0.96–0.97]; F1 0.83; AUPRC 0.46; Sensitivity 0.89; Specificity 0.87) (Supplemental Table 5). Similar results were found for the parameters measured on a per hour basis (HRV, HR, SpO2). A model including these parameters had an F1 score of 0.81 at 49-days before the flare event (49 days before flare; AUC 0.96 [0.96–0.96]; F1 0.81; AUPRC 0.49; Sensitivity 0.90; Specificity 0.86 (Supplemental Table 6).

DISCUSSION

The IBD Forecast study demonstrates the feasibility of identifying and predicting IBD flares using non-invasive commonly used wearable technologies. We observed that the physiological metrics collected from wearable devices are altered up to 7 weeks prior to the development of inflammatory and symptomatic flare periods. This is the first study to our knowledge, that demonstrates that wearable devices can identify the worsening of a chronic inflammatory disease as well as differentiate whether underlying inflammation is present in symptomatic and asymptomatic individuals.

IBD monitoring relies on cross-sectional and often invasive or inconvenient means to assess disease activity. This results in long periods during which no disease assessments occur. Furthermore, there is no convenient means to predict the development of inflammation or disease flares. The advancement of digital technologies provides an opportunity for frequent, passive, and real-time assessments of an individual's physiological status, filling in the data gaps generated by traditional disease assessments. The ability of wearable devices to passively collect data, coupled with their popularity, has generated increasing interest in their use to monitor health and disease.

Optical and actigraphy sensors on wearable devices can frequently measure several important physiological metrics. ANS dysfunction, characterized by an uncoupling of the ANS and hypothalamic pituitary adrenal axis, has been found in individuals with IBD, with greater degrees of dysregulation seen in individuals with active disease compared to remission.³⁶ Indirect measures of ANS activity, specifically HRV, have been cross sectionally correlated to IBD activity in several studies. IBD is associated with overall decreases in HRV indices, with this inverse relationship strongest for the PNS.¹⁵ Our group furthered these observations by demonstrating a longitudinal relationship between inflammation and SNS tone, and an inverse relationship between inflammation and parasympathetic tone.¹⁷ HR is similarly affected by the ANS through innervation of the sinoatrial node of the heart, with the SNS increasing.¹⁶ Thus, physiological metrics that are measured by wearable devices are altered by ANS dysfunction and the SNS activation seen with active inflammation in IBD. The ability of such metrics to identify and predict underlying inflammatory events has been demonstrated in studies using wearable devices for the identification and prediction of COVID-19 infections. Such studies demonstrated significant increases in RHR and mean HR and altered HRV circadian patterns during and before an infection.^{31,37–39}

We similarly observed significant changes in the physiological metrics measured by wearable devices during IBD flares. There were significant changes in HRV circadian

patterns which differentiate periods of inflammation from remission. The MESOR of the circadian pattern for SSDN was lower during flares, reflecting increased sympathetic tone. Furthermore, when evaluating the RMSSD metric, we note other changes in the HRV circadian pattern, demonstrating an overall alteration of ANS activity. An increase in mean HR and RHR were observed during periods of inflammation. Similar changes in all three physiological metrics were observed during symptomatic flares.^{27–29} Interestingly, we found that daily steps were lower during inflammatory flares, however we did not find a difference in steps with symptomatic flares. This observation is in line with one unpublished study demonstrating lower steps the week before inflammatory marker elevation.⁴⁰ While this warrants further evaluation, an interpretation of daily steps can be challenging as physical activity measurements are prone to bias due to device non-adherence and are not measuring a physiological parameter.

Beyond the identification of flares, we found that physiological metrics collected from wearable devices can identify inflammatory and clinical flares up to 7 weeks early. This observation highlights that measurable physiological changes occur before the development of flare events. Similar observations have been made with FC, which can rise 6 months before the development of clinical flares.⁴¹ This supports our observation that there can be a lag time between physiological status and flare. In the case of wearable-based metrics, this change preceding flares may similarly be secondary to the development of subclinical inflammation which alters ANS activity, and the parameters measured by wearable devices that it impacts. One limitation of our study design is that regular inflammatory assessments are not obtained precluding determination of when individuals transition from remission to flare. To address this limitation, our group is evaluating transitions between remission and flare states, and their impact on wearable based metrics, to further explore this finding.

The physiological source for many wearable-based biomarkers creates an opportunity to better characterize the disease status of individuals with IBD. There is discordance between symptoms and inflammation in IBD.^{42,43} We observed that HRV, HR, and RHR were able to differentiate the presence or absence of inflammation in symptomatic individuals. This analysis demonstrates that in IBD, inflammation is a primary driver of the changes in wearable measured physiological metrics. However, our observation that HR and RHR could differentiate symptomatic from asymptomatic periods, even in the absence of inflammation, shows that symptoms alone have some impact on physiological parameters. Further evaluation of this observation is needed to understand the degree and contribution of symptoms and inflammation on wearable-derived metrics. Interestingly we observed that differences in systemic oxygen levels were predictive of inflammatory and clinical disease flares. While the intestinal epithelium normally is in a state of relative hypoxia, which is exacerbated during active inflammation, there have not been descriptions of systemic oxygen levels measured during disease flares.⁴⁴ The pathophysiology explaining this observation is not clear and may be secondary to alterations of the oxygen saturation curve which result in slight changes in pulse oximetry measurements, which can be secondary to carbon dioxide concentrations, acid-base status, and temperature.⁴⁵

Our findings demonstrate the possibility of digitally collected physiological metrics to serve as a novel biomarker of disease activity. While there is no gold standard for detecting IBD

flares, monitoring using wearable devices may potentially be better given their continuous and physiological grounding. While the physiological changes we observed during periods of flare differed significantly compared to periods of remission, the absolute differences in these values were small. These small differences, however, are often sufficient for individual flare detection via machine learning and deep learning algorithms. This has been performed in other conditions with similar findings, resulting in a clinically relevant outcome.^{37,46} Other areas warranting further exploration include the utility of wearable biomarkers to identify and predict IBD medication response,⁴⁷ as well as their ability to serve as future endpoints in clinical trials.

There are several limitations to our study. First, the physiological metrics measured by the wearable devices are not specific and can be impacted by other factors, including intercurrent conditions. To mitigate this with regards to HRV, we controlled for the covariates of BMI, age and sex in our analysis. However, this is an important area that requires further study and validation. Additionally, inflammatory assessments were collected as part of standard of care evaluations. This limits our ability to determine when an individual transitions from inflammatory remission to inflammatory flare. To address this limitation, we imputed blood and stool results ± 7 days around each collection timepoint, to account for the fact that in IBD, inflammatory markers are elevated for extended periods of time around flares.⁴⁸ However, since we do not know when individuals transition from remission to inflammation, there is the potential for the misclassification of disease activity periods. Similarly, several potential definitions can be used to determine periods of symptomatic flare compared to symptomatic remission. This includes assumptions around the number of daily surveys that need to be answered per 7-day period, and the number of surveys consistent with symptomatic activity in a 7-day period needed to classify that period as “flare”. While we explored this in a sensitivity analysis, there are no well-defined definitions in the literature, therefore leading to the potential to misclassify symptomatic periods. Additionally, there is a limitation in our mixed-effect logistic regression models employed to predict flare-ups from physiological data. Overall, the study data set is imbalanced with a high prevalence of both inflammatory and symptomatic remission states compared to flare events. Furthermore, many flares occur within the same individuals. This low variability within patients can result in some biasing of the model’s outcome and AUC results. However, we settled on this approach as the data set contains repeated measures, with individuals contributing multiple outcomes (flares). An additional limitation is that the impact of medication is not controlled in the analyses. There is significant variation in medication type, dose, frequency and the timing of medication changes. There are variations in how the medications overlap with provided symptomatic and inflammatory assessments. This makes it challenging to control for medications in the analysis. Lastly, digital study cohorts have been shown to differ in comparison to non-digital study cohorts, including in composition and adherence to chronic medical management. While this is a limitation to all digital studies it is important to recognize the potential for bias or non-generalizability of results that this can introduce.

Conclusions

We demonstrate that physiological metrics collected longitudinally from wearable devices can identify and change before the development of inflammatory and symptomatic disease flares. Furthermore, physiological metrics can differentiate whether there is underlying inflammation present during symptomatic flares. These findings support the further evaluation of wearable devices in the monitoring of IBD.

Supplementary Material

Refer to Web version on PubMed Central for supplementary material.

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Data Availability:

Data collected as part of this study will be made available upon reasonable requests. Analytic methods, and study materials will not be made available to other researchers beyond what is described in the current study.

Abbreviations:

ANS	Autonomic nervous system
AUC	Area under the curve
AUPRC	Area under precision-recall curve
BMI	body mass index

BPM	beats per minute
CD	Crohn's disease
CI	confidence interval
CRP	C-reactive protein
FC	fecal calprotectin
HR	heart rate
HRV	heart rate variability
IBD	inflammatory bowel disease
IBI	inter-beat intervals
LRT	Likelihood Ratio Test
MESOR	midline statistic of rhythm or overall rhythm-adjusted mean
PNS	parasympathetic nervous system
PRO	patient reported outcome
RMSSD	root mean square of successive differences between normal heartbeats
RHR	resting heart rate
SDNN	standard deviation of the IBI of normal sinus beats
SEM	standard error of the mean
SNS	sympathetic nervous system
SpO2	oxygen saturation of arterial hemoglobin
UC	ulcerative colitis

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What you need to know:**Background and Context:**

Inflammatory bowel disease flares are challenging to predict and rely on cross sectional single time point assessments.

New Findings:

Wearable devices collect nearly continuous physiological metrics which can identify and predict inflammatory and symptomatic inflammatory bowel disease flares.

Limitations:

Laboratory measurements of disease activity were collected as part of participant's standard of care management, limiting the ability to determine precise transitions points between uninfamed and inflamed states.

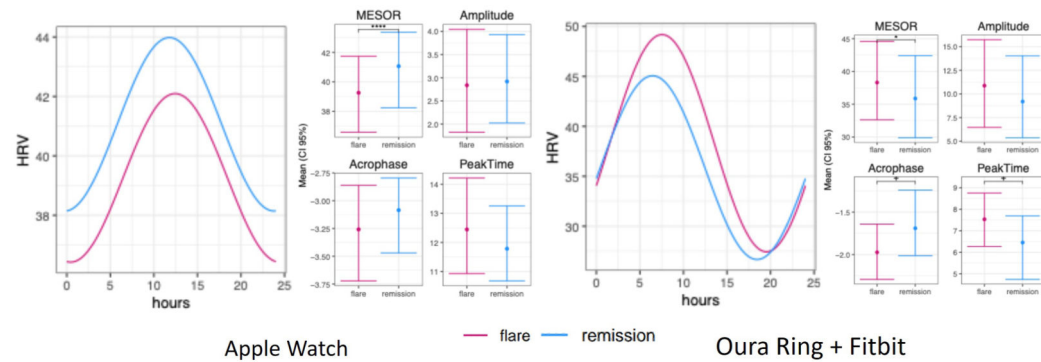
Clinical Research Relevance:

This is the first large study to demonstrate the physiological metrics collected from wearable devices can identify, differentiate, and predict inflammatory and symptomatic flares. This demonstrates the potential ability of wearable devices to be used in the monitoring of inflammatory conditions such as inflammatory bowel disease.

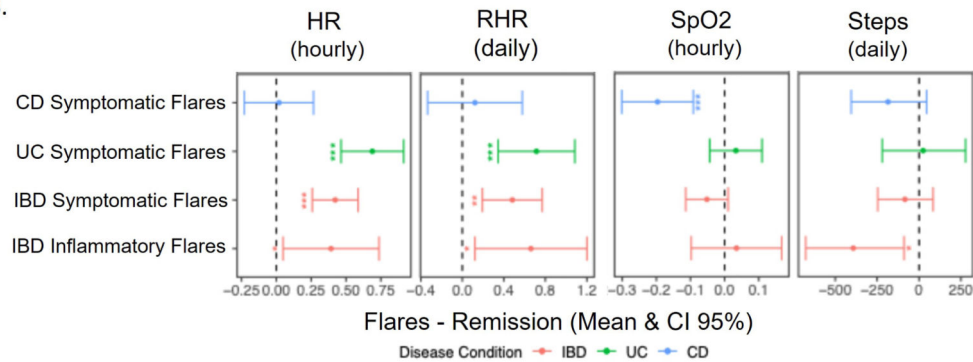
Basic Research Relevance:

The dense physiological assessments wearable devices collect can potentially be used to complement and extend physiological observations about inflammatory bowel disease.

A. Inflammatory Flares

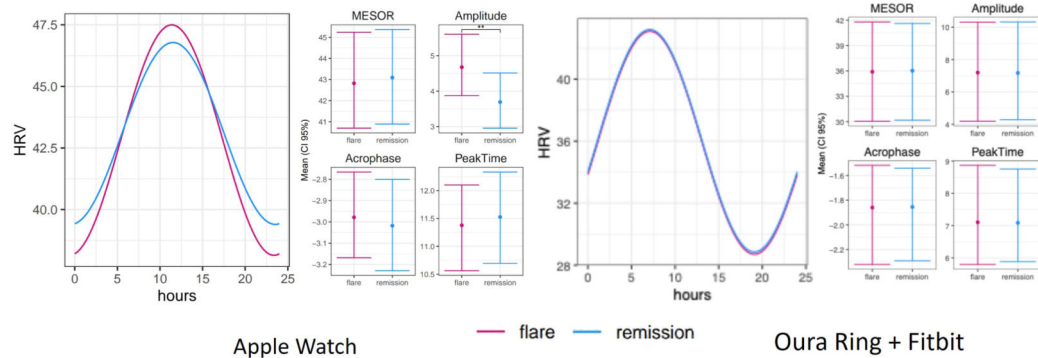


B.



Flares - Remission: Positive values indicate an increase during Flares

C. Symptomatic Flares

**Figure 1.**

The relationship between physiological metrics collected from wearable devices and inflammatory and symptomatic flares. (A) HRV circadian patterns during periods of inflammatory flares compared to periods of inflammatory remission with means and 95% CI for HRV measures. (B) The HR, RHR, SpO2, and daily steps during periods of symptomatic and inflammatory flares compared to periods of remission. (C) HRV circadian patterns during periods of symptomatic flares compared to periods of symptomatic remission with estimated marginal means and 95% CIs for HRV parameters. Stars represent the

comparison between Flare and Remission states. [†]P≤0.1, *P≤0.05, **P≤0.01, ***P≤0.001, ****P≤0.0001

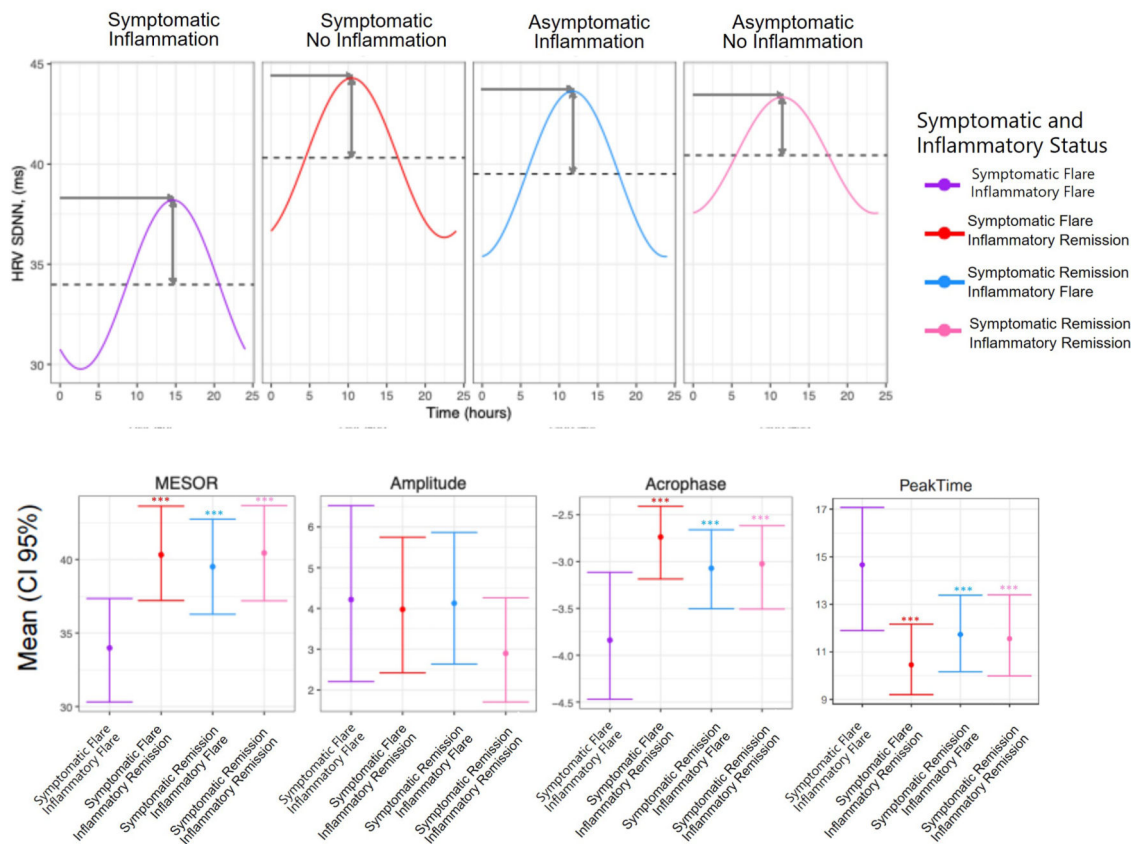
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A. Circadian Features of HRV Stratified by Symptomatic and Inflammatory Status



B.

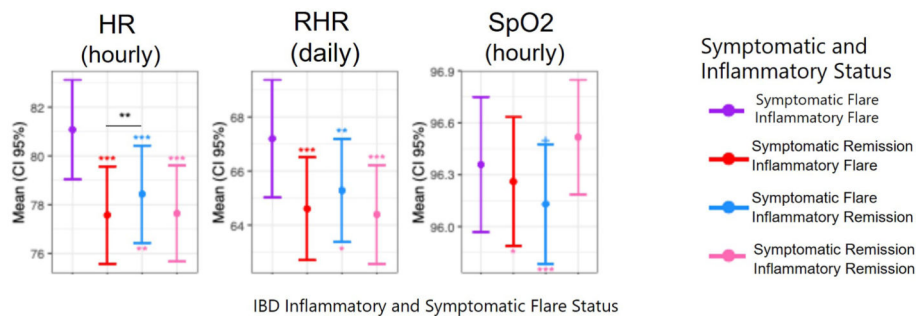


Figure 2. Analysis of physiological metrics by both inflammatory and symptom status in participants using the Apple Watch (HRV, SpO2) and all devices (HR, RHR). (A) The mean HRV circadian patterns for each interaction state is presented and the estimated marginal means and 95% CIs for HRV metrics measured during each state is shown (B) HR, RHR, and SPO2 metrics stratified based on each inflammatory and symptom status with estimated marginal means and 95% CIs for HRV measures. Stars atop the upper 95% CI are comparisons with the symptomatic and inflammatory flare (purple) group. Stars under

the lower 95% CI are comparisons with symptomatic and inflammatory remission (pink) group. Other comparisons are indicated in black. [†]P≤ 0.1, *P≤0.05, **P≤0.01, ***P≤0.001, ****P≤0.0001

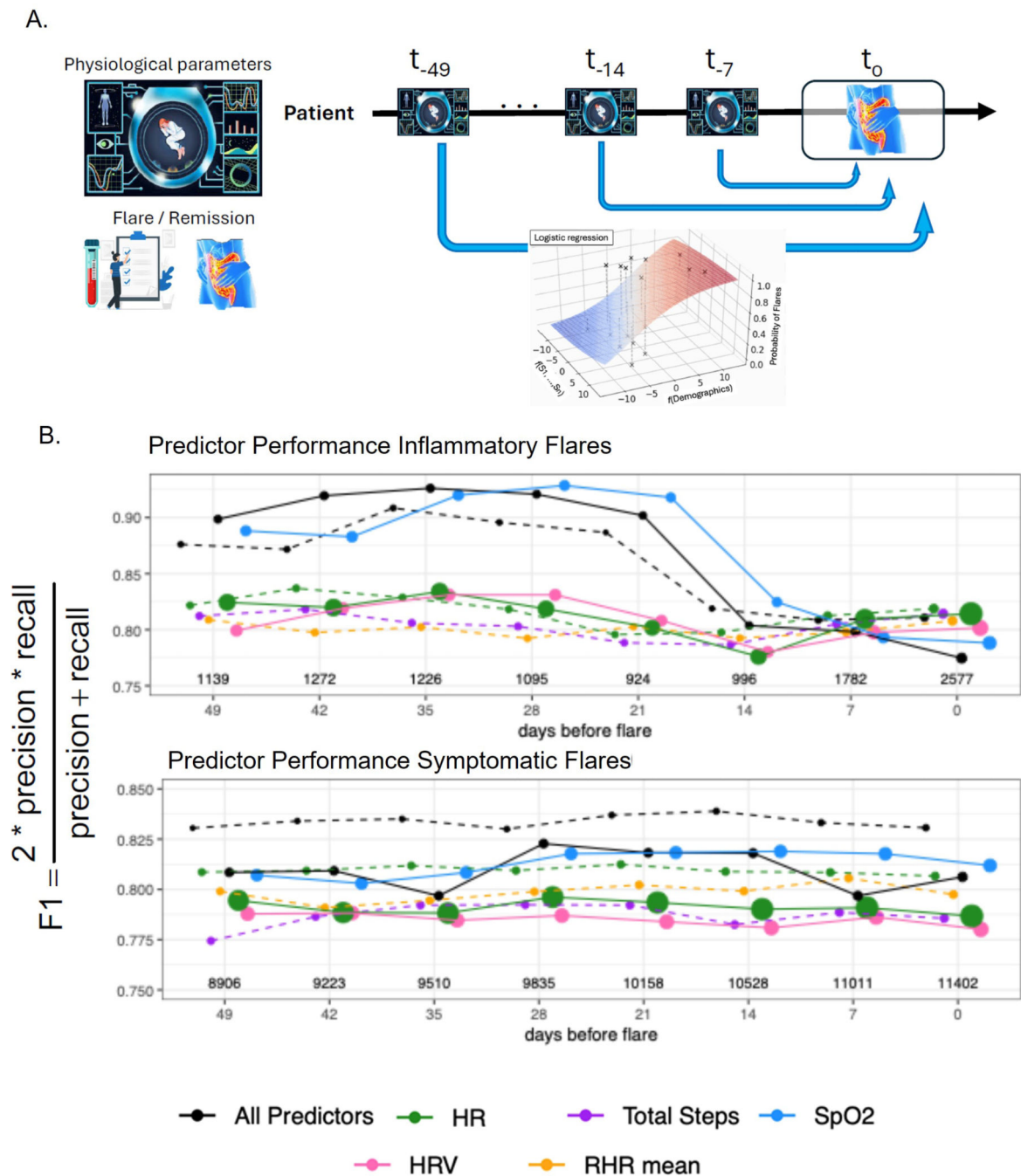


Figure 3.

(A) Association of early physiological metric (taken 7, 14, 21, 28, 35, 42, and 49 days before flare periods) with subsequent inflammatory and symptomatic flares up to 7 weeks before the events were assessed. (B) The F1 value summarizes the performance of the model that includes individual physiological metrics or a combination of them at each time point. The F1 score balances precision and recall and is optimal for evaluating imbalanced datasets. The F-score ranges from 0.0 to 1.0, representing perfect precision and recall. The number of physiological datapoints analyzed at each time point is listed at each time interval. Solid

lines represent physiological metrics analyzed on a per hour basis. Dashed lines represent metrics analyzed on a per day basis.

Table 1.

Demographic information for study participants.

	Participants with Crohn's Disease (N=196)	Participants with Ulcerative Colitis (N=113)	Overall Cohort (N=309)
Age, mean (SD)	39.6 (14.3)	40.5 (14.7)	39.9 (14.4)
Missing Values (%)	0 (0)	1 (0.9)	1 (0.3)
Sex, female (%)	134 (68.4)	74 (65.5)	208 (67.3)
BMI, mean (SD)	25.7 (5.51)	25.7 (5.63)	25.7 (5.54)
Missing Values (%)	10 (5.1)	9 (8.0)	19 (6.1)
Race			
Asian	7 (3.6%)	7 (6.2%)	14 (4.5%)
Black	3 (1.5%)	3 (2.7%)	6 (1.9%)
Native-American	2 (1.0%)	1 (0.9%)	3 (1.0%)
White	178 (90.8%)	100 (88.5%)	278 (90.0%)
Missing Variable	6 (3.1%)	2 (1.8%)	8 (2.6%)
Ethnicity			
Hispanic	11 (5.6%)	7 (6.2%)	18 (5.8%)
Not Hispanic	179 (91.3%)	101 (89.4%)	280 (90.6%)
Missing Variable	6 (3.1%)	5 (4.4%)	11 (3.6%)
Smoking Status			
Current	6 (3.1%)	3 (2.7%)	9 (2.9%)
Never	146 (74.5%)	91 (80.5%)	237 (76.7%)
Past Smoker	42 (21.4%)	19 (16.8%)	61 (19.7%)
Missing Variable	2 (1.0%)	0 (0%)	2 (0.6%)
Reported Medical Conditions			
Asthma	27 (13.8%)	8 (7.1%)	35 (11.3%)
History of Cancer	9 (4.6%)	3 (2.7%)	12 (3.9%)
Chronic Kidney Disease	2 (1.0%)	0 (0%)	2 (0.6%)
Chronic Lung Disease	2 (1.0%)	1 (0.9%)	3 (0.9%)
Diabetes Mellitus	3 (1.5%)	1 (0.9%)	4 (1.3%)
Hypertension	9 (4.6%)	5 (4.4%)	14 (4.5%)
No Medical Conditions	141 (71.9%)	91 (80.5%)	232 (75.1%)
History of IBD Related Surgery			
No	95 (48.5%)	109 (96.5%)	204 (66.0%)
Yes	97 (49.5%)	4 (3.5%)	101 (32.7%)
Missing Variable	4 (2.0%)	0 (0%)	4 (1.3%)
IBD Medications Use at Enrollment			
Mesalamines	15 (7.7%)	35 (31.0)	50 (16.2)
Corticosteroids	16 (8.2%)	19 (16.8)	35 (11.3)
Biologic Agents	129 (65.8)	37 (32.7)	166 (53.7)
Small Molecules	0 (0%)	6 (5.3%)	6 (1.9%)
Immune Modulators	17 (8.7)	5 (4.4)	22 (7.1)

	Participants with Crohn's Disease (N=196)	Participants with Ulcerative Colitis (N=113)	Overall Cohort (N=309)
No Medication Reported	8 (4.1%)	5 (4.4%)	13 (4.2%)
Crohn's Location			
L1	89 (45.4%)		
L2	25 (12.8%)		
L3	74 (37.8%)		
L4	47 (24.0%)		
Crohn's Behavior			
B1	39 (19.9%)		
B2	102 (52.0%)		
B3	55 (28.1%)		
Perianal Modifier	59 (30.1%)		
Ulcerative Colitis Location			
Left Sided Disease		39 (34.5%)	
Extensive Colitis		38 (33.6%)	
Proctitis		11 (9.7%)	